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EFY-CERTIFIED Skill DEVELOPMENT CENTRE

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4.5V to 35V, and output voltage can be adjusted from 1.2V to 33V; however, we need output voltage lower than 17V.

Output voltage V_{OUT1} can be calculated using the following relationship:

$$V_{OUT1} = 1.25V (1 + (VR2 + VR3)/R7))$$

Output voltage V_{OUT2} is around 1.5V lower than V_{OUT1} , and can consequently start from 0V.

Transistors T1 and T2 are implemented for adjustable current-limiting function along with potentiometer VR3. Minimum output current is around 0.35A, and depends on resistors R2 and VR3.

Wiper of VR3 should be at the right-most position to get minimum output current, and at the left-most position for maximum output current. Maximum output current is around 2A. When VR1 is adjusted for maximum output current, T1 and T2 will be on, and LED2 will glow. Otherwise, T1 and T2 will be off, and the LED2 will also be off.

Capacitors C4 and C9 prevent oscillations of T1 and T2 during transitional phases. Output voltage is adjusted with VR1 and

VR3. VR2 is used for coarse adjustment, while VR3 is used for more precise output voltage adjustment.

Construction and testing

An actual-size PCB layout for this power supply circuit is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on the designed PCB or veroboard. Connect around 18 to 20Vrms input to CON1. Glowing of LED1 indicates the presence of power supply in the circuit. LED2 glows when higher current is taken from the load. LED3 glows when outputs are available at CON2 and CON3.

Measure outputs across CON2 and CON3 using a voltmeter. You should be able to get output voltage V_{OUT1} from 1.2V up to around 16.5V, and V_{OUT2} from 0V to 15V depending on positions of VR2 and VR3. **EFY**

PARTS LIST

Semiconductors:

IC1	- LM350 adjustable voltage regulator
BR1	- 3A bridge rectifier
T1	- BC327 transistor
T2	- BC337 transistor
D1, D4	- 1N4007 rectifier diode
D2, D3	- 1N5408 rectifier diode
LED1-LED3	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

R1	- 100-ohm
R2	- 2-ohm, 3-watt
R3	- 300-ohm
R4, R5, R8, R9	- 2-kilo-ohm
R6	- 4.7-kilo-ohm
R7	- 180-ohm

Capacitors:

C1, C2, C6, C7, C10, C12	- 100nF ceramic disk
C3, C5	- 2200 μ F, 35V electrolytic
C4, C9	- 330pF ceramic disk
C8, C11	- 100 μ F, 35V electrolytic

Miscellaneous:

CON1-CON3	- 2-pin terminal connector
F1	- 2A fuse with holder
	- Heat-sink for LM350
	- 18V AC/DC power supply

(All these are available from kitsnspares.com)



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